Unsupervised Domain Adaptation for Building Extraction of High-Resolution Remote Sensing Imagery Based on Decoupling Style and Semantic Features

Jie Chen, *Member, IEEE*, Jingru Zhu, Peien He, Ya Guo, Liang Hong, Yin Yang, Min Deng, *Member, IEEE*, and Geng Sun

***Abstract*— When the buildings themselves or the environments they are located in change, it poses a cross-domain problem for building extraction. Since the semantics of the same category should be consistent across domains, a memory mechanism can be used to drive models to capture it. A key prerequisite for a satisfactory memory mechanism is that the memory content is highly task-relevant, i.e., the memory mechanism represents domain-invariant semantic features of buildings as much as possible. It is worth noting that images from different domains have diverse styles, which can interfere with the representation of domain-invariant semantic features of buildings. To maximize the effectiveness of memory mechanisms in cross-domain building extraction tasks, it is necessary to reduce the interference of style information. Therefore, we propose a cross-domain extraction method for buildings based on decoupling style-semantic features. It expresses the image style features and building semantic features separately and optimizes the domain-invariant features of buildings by constraining their relationship through an orthogonal loss. Specifically, on the one hand, style features within the same domain as well as semantic features within and across domains are narrowed down through contrastive learning to ensure the consistency of style features extracted within the same domain and the consistency of semantics across domains. On the other hand, the style features and domain-invariant semantic features in the memory mechanism can be stored and reused to fully exploit self-supervised information. Results of the cross-domain experiments show that the proposed method can achieve optimal building extraction.**

***Index Terms*— Building extraction, contrastive learning, domain adaption, invariant semantic features, memory bank, semantic segmentation.**

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Jie Chen, Jingru Zhu, Peien He, Ya Guo, Min Deng, and Geng Sun are with the School of Geosciences and Info-Physics, Central South University, Changsha, Hunan 410083, China (e-mail: sungeng@csu.edu.cn).

Liang Hong is with the College of Tourism and Geography Science, Yunnan Normal University, Kunming, Yunnan 650500, China (e-mail: hongliang20433@hotmail.com).

Yin Yang is with the School of Mathematics and Computational Science and the National Center for Applied Mathematics, Xiangtan University, Hunan 411105, China, and also with Hunan International Scientific and Technological Innovation Cooperation Base of Computational Science, Xiangtan, Hunan 411105, China (e-mail: yangyinxtu@xtu.edu.cn).

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I. Introduction

BUILDINGS are crucial elements in the urbanization process. High-resolution remote sensing images enable the investigation and analysis of building density, types, and distribution. Therefore, the utilization of high-resolution remote sensing images holds significant value in various fields [1], [2], [3]. Previous research works mainly relied on manually designed features such as texture [4], shape [5], and color [6] to extract visual characteristics for building extraction. However, with the increase in spatial resolution, remote sensing images become more detailed, and different objects may exhibit similar feature attributes. Moreover, variations in buildings themselves can lead to deviations in their features. In recent years, deep convolutional neural networks (CNNs) have been widely applied in various fields due to their powerful feature representation capabilities. Unlike traditional machine learning algorithms, CNNs can directly obtain deep features from data in an end-to-end manner and have been used to solve various visual-related tasks, such as lesion detection [7], autonomous driving [8], and remote sensing image interpretation [9]. Especially in the field of remote sensing, CNN-based methods have gradually become the mainstream techniques for change detection [10], object recognition [11], land cover classification [12], remote sensing target detection [13], and other remote sensing-related tasks. It is worth noting that CNN-based methods require large amounts of labeled data to support model training.

However, buildings undergo stylistic and material changes with the process of urbanization, resulting in domain differences between buildings from different domains, which can lead to poor model performance and insufficient generalization ability. In the field of computer vision, unsupervised domain adaptation (UDA) methods have been used to tackle domain differences. Currently, many UDA methods have been applied to building extraction. For example, Shi et al. [14] designed an image style transfer network to transfer the style of the

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images based on a cycle-consistent generative adversarial network (CycleGAN) and then performed building extraction on the transformed images. Compared to natural images, high-resolution remote sensing images have more complex backgrounds and large domain differences, making it difficult for style transfer methods to optimize complex loss functions and meet the requirements of knowledge transfer. Recently, some researchers focused on reducing domain differences at the feature level by continuously minimizing the feature distance between the source and target domain, thus enabling the model to extract domain-invariant features [15]. Domain-invariant features can represent the essential semantics of a certain class of objects and provide good representations of that class across different domains, while style is domain-specific and nontransferable information. Nevertheless, the features extracted by this model are often a mixture of semantic and style information, and do not eliminate the negative impact of style on domain-invariant features. Intuitively, objects of the same category should exhibit semantic consistency across domains, and exploiting this consistency can drive the model to capture domain-invariant features of similar objects in different domains. Additionally, humans can utilize the memory mechanism of the brain to extract useful information from past experiences, store it, and reuse it to aid in the perception and understanding of current objects [16]. In the task of cross-domain building extraction, the building objects appearing in different domain images should be semantically consistent. Memory mechanisms can be utilized to construct memory banks to store and reuse domain-invariant building semantic features that can transfer critical knowledge in the cross-domain task. However, for the memory mechanism to achieve satisfactory results, the correspondence between the memory and the task should be sufficiently high. In the task of cross-domain building extraction, memory should consist of domain-invariant building semantic features. Therefore, to maximize the effectiveness of the memory mechanism in cross-domain building extraction tasks, it is necessary to more fully explore the domain-invariant building features to improve the accuracy and generalization of the model. Contrastive learning [17], [18] extract useful information from a large number of unlabeled samples and assist various downstream tasks. It aims to bring similar or augmented samples as close as possible in the feature embedding space while pushing different-class samples as far apart as possible. Contrastive learning in the feature space can be utilized to obtain accurate and memorable domain-invariant semantic features of buildings.

Based on the above analysis, we propose a UDA network with decoupling style and semantic features (DSSFNet) for cross-domain building extraction. It utilizes the semantic consistency of buildings and the style consistency within the same domain of images for cross-domain building extraction. The method first uses independent style and semantic encoders to express style and semantic features, respectively. Next, inspired by the feature decoupling works [19], [20], [21], we impose a linearly independent orthogonal constraint between the two types of features to guide the style and semantic features to be mutually independent and decoupled. Furthermore, based on the semantic consistency between building domains and the style consistency within the image domain, a memory feature optimization module is constructed. Through the memory mechanism and contrastive learning, the model achieves an accurate representation of domain-invariant features. Specifically, a memory bank is constructed for both style and semantic features, and the model is optimized using both semantic contrastive loss and style contrastive loss. In brief, our main contributions are presented as follows:

1) A cross-domain building extraction network, named DSSFNet, is proposed to achieve a deeper optimization of domain-invariant semantic features of buildings by utilizing the domain-specific image style information within the domain.
2) The decoupling of style-semantic features is designed, wherein style features are utilized to facilitate the accurate representation of semantic features with linearly independent orthogonal constraints.
3) Memory mechanism and contrastive learning strategy are employed to constrain the consistency of domain-specific style information and the consistency of interdomain semantic information, thereby reducing the differences in style features within the domain and the variances in semantic features among similar buildings.

The remainder of this article is organized as follows. Section II describes the related work. Section III illustrates our proposed method. Section IV presents the experimental results, and Section V concludes our work.

II. Related Works

This section illustrates the related work about building extraction, including the four following parts: deep learning-based building extraction, UDA in the remote sensing field, memory bank, and contrastive learning.

A. Deep Learning-Based Building Extraction

Compared to traditional machine learning algorithms, CNN eliminates the need for manual feature engineering. It can automatically extract deep features from data and use them for various downstream tasks. With the in-depth research on various techniques such as attention mechanisms, co-occurrence visual patterns, and multiscale fusion strategies, an increasing number of semantic segmentation methods based on fully convolutional networks (FCN) [22] have been proposed, such as U-Net [23], Deeplab series [24], [25], [26], [27], PSPNet [28], SENet [29]. Building extraction is a binary semantic segmentation task in remote sensing image interpretation, and many researchers have used FCN techniques for building extraction in remote sensing images. For example, Chen et al. [30] proposed a dense residual neural network that reduces the number of network parameters without compromising accuracy, thereby further improving the efficiency of building extraction. Huang et al. [31] introduced a lightweight building extraction network that fuses deep and shallow features and uses squeeze-and-excitation modules to weight the feature channels. Other researchers have emphasized extracting more accurate building features. For instance,

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Liu et al. [32] designed dense boundary blocks based on dense blocks, boundary blocks, and spatial channel attention blocks, addressing the issues of boundary blurriness and intraclass inconsistency, thereby improving the accuracy of building extraction based on U-Net. Shao et al. [33] designed and optimized prediction modules and residual refinement modules to enhance the accuracy of building extraction. Deng et al. [34] proposed an encoder-decoder network based on grid attention gates and atrous spatial pyramid pooling, gradually optimizing and extracting building features. An uncertainty-aware network, He et al. [35] proposed to alleviate the uncertain prediction problem in building extraction. Furthermore, some studies have utilized auxiliary information in the images. For example, Guo et al. [36] improved the interpretation of remote sensing images by leveraging scene priors. They performed multilabel scene classification on input images and mapped buildings in different scenes at the pixel level. Recently, inspired by the success of Transformer [37] in computer vision, Transformer-based methods have gradually been applied to building extraction and have shown excellent performance in extracting long-range dependencies of geo-objects and the global context of buildings. For example, a BuildFormer, Wang et al. [38] is proposed to capture the global context while preserving spatial-detailed features of buildings. PolyBuilding, Hu et al. [39] proposed a transformer-based polygonal building extraction approach to simultaneously predict the bounding boxes and polygons for the building instances.

However, these methods rely heavily on large amounts of labeled data. When models trained on one dataset are tested on another dataset, it is often challenging to achieve satisfactory building extraction results due to domain differences. Therefore, in order to maintain good performance on data from different domains and to meet the requirements of cross-dataset building extraction applications, it is necessary to conduct research on cross-domain building extraction in high-resolution remote sensing images.

B. UDA in Remote Sensing Field

In the field of remote sensing, the visual presentation of remote sensing images is often influenced by various conditions such as lighting, viewing angle, soil moisture, and topography. These differences affect the spectral characteristics of observed objects, which in turn affects the distribution of information categories in the feature space [40]. As a result, models trained on labeled data cannot be directly applied to handle unlabeled new remote-sensing images from different domains. Therefore, UDA has gradually been employed to address cross-domain remote sensing image interpretation tasks, including cross-domain building extraction.

In the field of remote sensing, the application of UDA can be traced back to 2001, when Bruzzone and Prieto [41] proposed an unsupervised maximum likelihood classification method for land cover classification. Based on this, in 2002, the same team cascaded the maximum likelihood classifier with a radial basis function neural network to form an unsupervised classification system, which achieved good cross-domain land cover classification results on multitemporal and multisource datasets [42]. In 2009, the research team combined UDA with support vector machines and introduced the domain adaptation support vector machine, which proved its effectiveness on both remote sensing and natural image datasets [43], [44]. With the rise of deep convolutional networks in the field of computer vision, the fusion of UDA with deep learning has begun. In the field of remote sensing, Romero et al. [45] proposed the first UDA deep learning method for remote sensing image classification. Since then, deep learning-based methods have gradually become the mainstream approach to address domain adaptation challenges in remote sensing.

Generally, there are two main approaches to conducting remote sensing domain adaptation tasks based on deep learning.

1) Image-Level Alignment: The core of image-level alignment methods for remote sensing domain adaptation tasks is to transform and align the style of two datasets, and then perform the corresponding downstream tasks. For example, Bejiga and Melgani [46] used generative adversarial networks (GANs) to transform images for unsupervised remote sensing image classification. Benjdira et al. [47] proposed the first method for cross-domain semantic segmentation of remote sensing images using GANs, which reduced the domain differences by transforming the source-domain image into the target-domain image and performing semantic segmentation on target domain images. Tasar et al. [48] improved cross-domain semantic segmentation accuracy by constructing a color mapping GAN, which only involved one element-wise matrix multiplication and one matrix addition operation to convert the color of source domain remote sensing images to the color of target domain images. StandardGAN normalized each source and target domain to have similar data distributions and then trained a classifier on the normalized source domain and performed segmentation on the normalized target domain [49]. Vega et al. [50] proposed the first CycleGAN-based UDA method for change detection in remote sensing images, which preserved the semantic structures and category transformations in adaptive images. However, it should be noted that due to the more complex backgrounds in high-resolution remote sensing images compared to natural images, the optimization of complex loss functions in GAN-based image-level style transfer methods are challenging, making it difficult to train a stable network and achieve satisfactory knowledge transfer.

2) Feature-Level Alignment: Another approach for remote sensing domain adaptation tasks is feature-level alignment methods, which generally align the target domain features with the source domain features in the feature space. For example, Xie et al. [51] trained a transferable FCN model to predict nighttime lighting based on daytime images, enabling mapping of poverty areas worldwide. Othman et al. [52] optimized the network by jointly minimizing three regularization terms: source domain cross-entropy loss, the maximum average discrepancy between the source and target domain distributions, and target domain geometry, thereby achieving unsupervised cross-domain scene classification. Liu et al. [53] measured the differences in edge and conditional distributions of each class and narrowed the feature sets of the source and target domains

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in the reproducing kernel Hilbert space for cross-domain multimodal remote sensing image classification. Koga et al. [54] proposed a method combining feature correlation alignment with adversarial learning for cross-domain vehicle detection in remote sensing images. Zhang et al. [55] balanced the impact of the source and target domains on remote sensing image scene classification by maximizing subspace correlation and aligning dynamic statistical distributions. Zhang et al. [56] used adversarial learning to align the target domain features with the source domain features in the feature space and gradually improved road recognition accuracy on the target domain by applying a self-training strategy. Zhu et al. [57] introduced a novel UDA semantic segmentation network for high-resolution satellite image segmentation by embedding a domain-invariant prototype feature memory module. It can be observed that the existing feature-level domain adaptation methods mostly focus on optimizing domain-invariant features at the semantic level, while neglecting the impact of image-level domain style information. However, style and semantic information are often intertwined, and ignoring domain style information makes it difficult to obtain accurate domain-invariant building semantic information.

C. Memory Bank

Auxiliary memory units can be incorporated into a memory bank to store and reuse task-related information, thereby enhancing network capacity. Neural Turing machines [58] achieve this by utilizing an external memory module, where the controller uses a write head to input information and a read head to retrieve information from the memory module. This allows for increased memory capacity and flexible access to stored data. Weston et al. [59] were the first to use memory components to store scene information, thus enabling long-term memory capabilities. Sukhbaatar et al. [60] proposed an end-to-end model that can be used as a continuous form of a memory network. Currently, there are some methods that utilize memory mechanisms to address domain adaptation problems. For example, MemoryAdaptNet [57] established a domain-invariant feature memory bank and enhanced the model's semantic expression capabilities on the target domain through feature fusion. Although this method utilizes memory mechanisms to leverage domain-invariant features, they do not explore the accuracy of domain-invariant features in depth. The obtained domain-invariant features are actually pseudo-domain-invariant features that are influenced by style information, leading to limited model capabilities. Therefore, the key to making memory mechanisms effective is to obtain accurate domain-invariant features.

D. Contrastive Learning

Contrastive learning, as an important member of self-supervised learning, has been widely applied in computer vision, natural language processing, and other fields. Its objective is to maximize the proximity of augmented samples in the embedding space while maximizing the dissimilarity between different samples. In contrastive learning, the model is typically constrained using the InfoNCE loss function [17]. The InfoNCE loss is a contrastive loss function defined as follows:

$$L_q = -\log \frac{\exp(q \cdot k_+/\tau)}{\sum_{i=0}^{k} \exp(q \cdot k_i/\tau)} \tag{1}$$

where the numerator represents the similarity of positive pairs, and the denominator represents the overall similarity of positive and negative pairs. Minimizing the loss requires maximizing the numerator and minimizing the denominator, which increases the similarity of positive pairs and decreases the similarity of negative pairs. τ is a temperature coefficient that adjusts the focus on challenging samples.

Contrastive learning has achieved significant success in computer vision tasks through unsupervised contrastive pretraining, especially when the number of negative samples is large, as more negative samples are likely to cover the underlying data distribution. However, the quantity of negative samples is always limited by the batch size. Recently, researchers proposed a momentum contrast method to address the issue of insufficient negative samples within a single iteration by maintaining a large queue of negative samples. Specifically, this method updates the negative sample encoder using periodic momentum updates instead of backpropagation [18]. SimCLR introduces nonlinear transformations and adversarial loss functions to improve the quality of learnable representations [17]. BYOL does not use negative samples but applies parameter freezing to the encoder and uses a momentum approach for parameter updates [61]. In addition to not using negative pairs, Simsiam also proposes a "stop-gradient" strategy to learn meaningful representations without requiring large batches of negative samples and a momentum encoder [62]. For cross-domain visual tasks, contrastive learning can be utilized to measure the distance between target domain features and domain-invariant features, thereby optimizing domain-invariant features. For instance, MDANet constructs a memory contrast module that aligns domain-invariant features of buildings with target domain-building features using contrastive learning and continuously updates domain-invariant features using a memory mechanism [16]. This method considers the semantic consistency of similar objects through memory mechanisms and optimizes domain-invariant features through contrastive learning. However, it overlooks the within-domain style consistency, leading to the interference of style information on semantic information representation, leaving room for further improvement in obtained domain-invariant features.

III. Methodology

A. DSSFNet Architecture

The DSSFNet framework, as shown in Fig. 1, consists of three main components: the domain style and building semantic encoder, the domain style and building semantic feature decoupling module, and the memory feature optimization module. These components respectively address domain style and building semantic feature representation, domain style and building semantic feature decoupling, and memory feature optimization. The source domain data can be defined as D_s,

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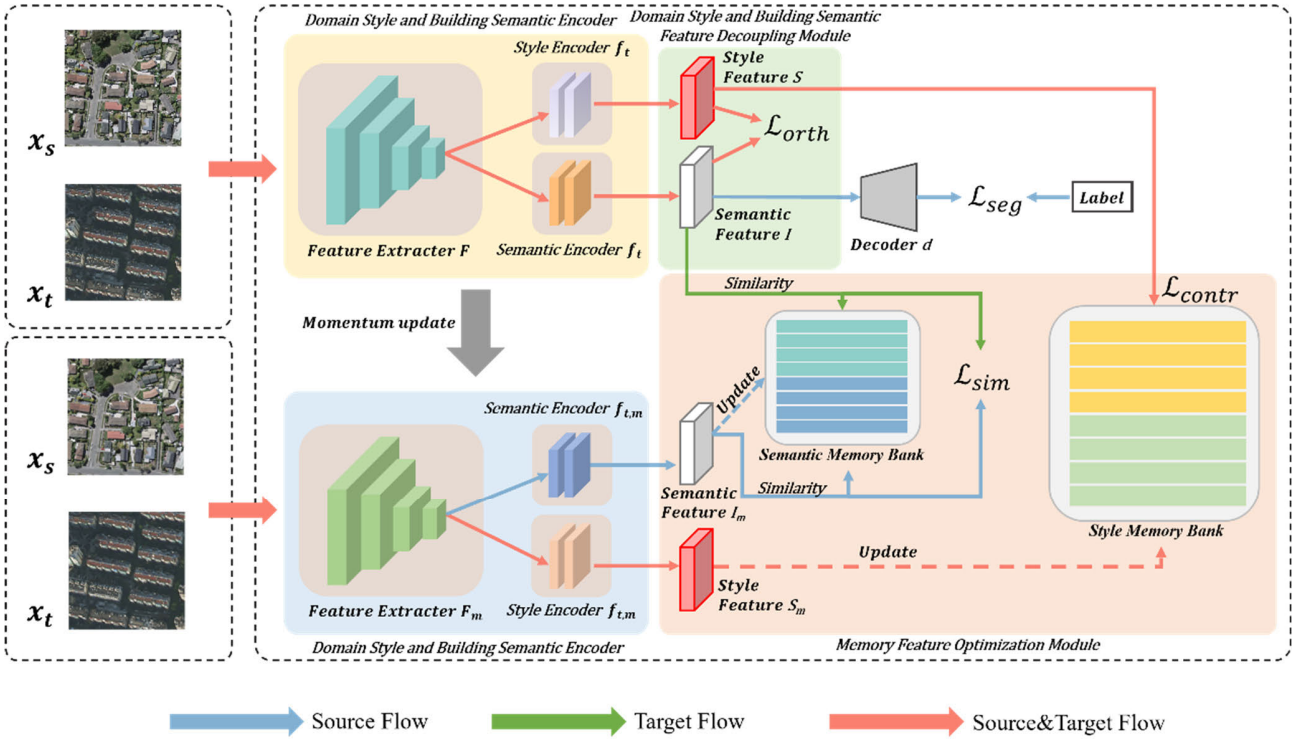


Fig. 1. Overview of DSSFNet.

which consists of images x_s and labels y_s. The target domain data, denoted as D_t, only include images x_t.

1) *Domain Style and Building Semantic Feature Representation.* To separate the style and semantic features of buildings from the original features, two independent feature encoders, the style encoder f_s and the semantic encoder f_t, are constructed to represent the style features S and the semantic features I, respectively.
2) *Domain Style and Building Semantic Feature Decoupling.* To ensure the independence of style and semantic features, a domain style and building semantic feature decoupling module is designed based on the underlying relationship between style and semantic features in the feature space. It utilizes an orthogonal constraint loss function $\mathcal{L}_{\text{orth}}$ to decouple the style and semantic features. Additionally, the style features are used to optimize the domain-invariant semantic features.
3) *Memory Feature Optimization.* Memory feature optimization aims to promote the accurate representation of domain-invariant building features by leveraging style consistency and semantic consistency constraints.

For supervised source domain data D_s consisting of images x_s and labels y_s, the semantic features I can be input to a decoder d to obtain semantic segmentation results. The segmentation results and building labels are used to compute binary cross-entropy loss $\mathcal{L}_{\text{seg}}$, ensuring that the model captures the semantic information of buildings. Semantic features should maintain semantic consistency within the same class, while style features should exhibit style consistency within the domain. To maintain feature consistency during long-term updates, a set of transfer models, F_m, $f_{s,m}$, and $f_{t,m}$, are obtained by applying momentum updates to the feature extractor F, style encoder f_s, and semantic encoder f_t, respectively. To optimize the transferable building semantic features and effectively utilize domain-style features, a memory optimization module is designed. It consists of a semantic memory bank and a style memory bank. The semantic memory bank is updated by inputting source domain features to $f_{t,m}$ to obtain semantic features I_m. The target domain features are input to f_t to obtain semantic features I. The semantic features should be consistent with the features in the semantic memory bank. Thus, the similarity between I_m and I with the features in the semantic memory bank is calculated. When the similarity is high, it indicates that the model captures the same semantic features and a semantic contrastive loss $\mathcal{L}_{\text{sim}}$ is designed to encourage a more accurate representation of domain-invariant semantic features. The style memory bank is updated by obtaining style features S_m from $f_{s,m}$ for different domains. Since style features from the same domain should have high similarity, a style contrastive loss $\mathcal{L}_{\text{contr}}$ is designed to penalize the inconsistency between the style features from the style memory bank and those obtained from f_s. The optimization of each component is achieved by applying the corresponding loss functions, encouraging the style encoder to focus more on the style information of the data and the semantic encoder to focus more on the semantic information of the buildings.

B. Domain Style and Building Semantic Feature Expression

The features generated by the feature extractor contain a significant amount of noise introduced by the domain-specific style information. This noise can negatively affect the semantic representation of buildings in cross-domain tasks, making

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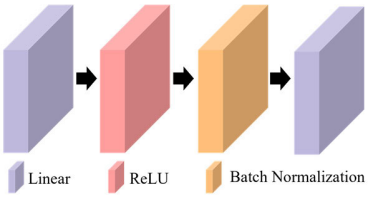


Fig. 2. Architecture of style encoder and semantic encoder.

it challenging for the model to transfer from the source domain to the target domain. Existing methods often focus on enhancing the representation of semantic features to align different domain features and obtain domain-invariant features. However, accurate domain-invariant semantic features still cannot be obtained due to ignoring style information. To address the aforementioned issues and separate the domain style from the building semantic features in the original features, this module constructs two encoders to express the style features and semantic features independently. Both the style encoder and the semantic encoder have the same network structure, which ensures equal extraction of semantic and style information from remote sensing images while maintaining their dimensional consistency, thus facilitating the calculation of orthogonal constraint loss and ensuring the separation and independence of style and semantic features of the image. As shown in Fig. 2, each encoder consists of a linear layer, a batch normalization layer, an activation function layer (ReLU), and another linear layer. Specifically, the features output by F are first fed into a linear layer, which applies a linear transformation to map the original features to a higher-dimensional space. Then, the batch normalization layer is used to enhance the communication and reuse of feature functionalities. The activation function layer reduces the interdependence between parameters and computational complexity. Finally, the linear layer is applied to transform the feature space dimensions to complete the transformation from the original features to either semantic features or style features.

C. Decoupling Domain Style and Building Semantic Features

It is evident that the style encoder and semantic encoder in the model structure achieve separate extraction of style features and semantic features. During the backpropagation process of the model, different loss functions can be used to control the parameter updates of the style encoder and semantic encoder, ensuring that the style encoder focuses only on the style information within the domain, while the semantic encoder captures only the semantic information of buildings. In this way, the two encoders can update their parameters without interfering with each other. However, since the relationship between style and semantic features is not considered, there is no guarantee that these two features are independent of each other in the feature space. Therefore, additional constraints are needed to decouple them from the feature space.

In linear geometry, sets of vectors that are orthogonal to each other are called orthogonal vector sets, while orthogonal sets of nonzero vectors are always linearly independent. Based on this deduction, when the style features S and the semantic features I are orthogonal, they are independent of each other. For this reason, we employ an orthogonality constraint to optimize the semantic features in the direction perpendicular to the style features, thus ensuring the independence of the two types of features.

D. Memory Optimization Module

Based on the memory mechanism and contrastive learning, this article establishes a memory optimization module to enhance the representation of domain-invariant semantic features of buildings by simultaneously constraining semantic consistency and domain style consistency.

In the constraint of building semantic consistency, for target domain data x_t, this article first maps the semantic features I_t obtained through the original semantic encoder f_t. Specifically, a multilayer perceptron (MLP) is used to map the features to the feature space of contrastive learning. Simultaneously, the decoder d generates pseudo-labels for target domain images. The pseudo-labels are resized to the same size as the projected semantic features, and multiple building feature vectors $V = v$ are obtained at the corresponding positions of the semantic features based on the pixel categories of the pseudo-labels. For source domain data, reliable supervision information is utilized to construct the building semantic feature memory bank V^I. However, due to the loss of model consistency during backward propagation updates, resulting in significant feature differences, it is difficult to effectively establish a memory bank that stores domain-invariant features. In this article, the original model is updated with momentum in order to obtain a set of transfer learning models with consistent structure but different parameters, gradually transferring the capability of the original model. Therefore, when constructing the semantic memory bank, x_s is first passed through the transferred semantic encoder $f_{t,m}$ to obtain the semantic features I_m. The features are projected in the same way as the target domain data and the building feature vectors are selected based on the source domain labels y_s. To obtain reliable building feature vectors, the decoder d_m is used to predict I_m. A certain confidence threshold is set, and the feature vectors are filtered based on the predicted probability map, resulting in feature vectors $V_m = v_m$ that meet the confidence conditions. The feature vectors are sorted according to confidence and the top K feature vectors are selected for memory storage. Since the capacity of the memory bank is limited, a first-in-first-out strategy is adopted to update the semantic memory bank V^I. The semantic features of building objects should exhibit consistency, both within and across the same domains. Therefore, for v_m and v, all feature vectors in the semantic memory bank V^I should be similar to them. To quantify the degree of similarity for use in the contrastive learning strategy, this article calculates the similarities p_m and p between v_m, v, and all features in V^I, respectively,

$$p_m^i = \frac{\exp\left(\langle v_m, v_i^I \rangle / \tau\right)}{\sum_{\mathbf{v}^{\mathbf{I}} \in} V^I \exp\left(\langle v_m, v^I \rangle / \tau\right)} \tag{2}$$

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where $V^I = [v_1^I, \ldots\ldots, v_i^I]$, τ is the temperature coefficient, $\langle \cdot, \cdot \rangle$ represents the cosine similarity between two parameters. For q memory features

$$p_m = \left[p_m^1, \ldots\ldots, p_m^q\right]. \tag{3}$$

Similarly, for p

$$p^i = \frac{\exp\left(\langle v, v_i^I \rangle / \tau\right)}{\sum_{\mathbf{v^I}\in} V^I \exp\left(\langle v, v^I \rangle / \tau\right)}. \tag{4}$$

For q memory features

$$p = \left[p^1, \ldots\ldots, p^q\right]. \tag{5}$$

In the domain-specific style consistency constraint, it is necessary to establish a style feature memory bank, denoted as S^S, containing sub-banks for both the source domain and the target domain. Following the same approach as the semantic consistency constraint, this part still utilizes the style features S_m generated by the transfer model, and the feature mapping is applied to construct the style features for building S^S, denoted as $S_m = s_m$. Since the domain of an image is known, the style features s_m can be divided into different domains, resulting in the source domain style sub-bank S_s^S and the target domain style sub-bank S_t^S. Similarly, the style features $S = [s_s, s_t]$ can be obtained through the original model. It is evident that the style features should be consistent within the same domain while exhibiting differences across different domains. Therefore, it is necessary to measure the similarity between S and the style features s_s^S in S_s^S, as well as the similarity between S and the style features s_t^S in S_t^S. Let $d = [s, t]$, where s represents the source domain and t represents the target domain. The similarity between style features within the same domain can be expressed as

$$\exp\left(\langle s_d, s_{d,i}^S \rangle / \tau\right) \tag{6}$$

where $S_d^S = [s_{d,1}^S, \ldots, s_{d,i}^S]$. The similarity between style features from different domains can be expressed as follows:

$$\exp\left(\langle s_d, s_{d',i}^S \rangle / \tau\right), d \neq d'. \tag{7}$$

E. Loss Functions

1) Segmentation Loss: In cross-domain building extraction tasks, only the source domain images provide reliable fully supervised information through labels. To ensure that the model captures the semantic information of buildings, this method utilizes the source domain data to design a binary cross-entropy segmentation loss function

$$\mathcal{L}_{\text{Seg}} = -\frac{1}{N_s} \sum_{i=1}^{N_s} y_s \cdot \log\left(\hat{y}_s\right) + (1 - y_s) \cdot \log\left(1 - \hat{y}_s\right) \tag{8}$$

where $\hat{y}_s$ represents the prediction results of the original model on the source domain, and N_s represents the number of source domain images in each batch.

2) Orthogonal Constraint Loss: This loss function utilizes the property of linear independence in an orthogonal nonzero vector set to induce the decoupling of semantic features and style features. The orthogonal constraint loss function L_{orth} can be represented in the form of L2 norm

$$\mathcal{L}_{\text{orth}} = \left\| I^T S \right\|_2 \tag{9}$$

where I^T represents the transpose of the semantic features and S represents the style features. By effectively establishing the latent relationship between style features and semantic features in this way, it is possible to decouple the style and semantic features on one hand, and on the other hand, implicitly optimize the semantic features with the assistance of style features.

3) Semantic Contrastive Loss: As described in (3)–(5), the similarities p_m and p are computed. The semantic contrastive loss function is defined as follows:

$$\mathcal{L}_{\text{sim}} = -\frac{1}{N} \sum p_m \cdot \log(p) \tag{10}$$

where N represents the number of feature pairs involved in the current batch.

The original contrastive loss function can only calculate the contrastive loss between v_m and v^i, as well as between v and v^i separately, without directly measuring the relationship between v_m and v. It is evident that if v_m and v are similar, the similarity scores of both v_m and v^i with v^i should be high. Based on this observation, a semantic contrastive loss function, as shown in (10), is designed to establish a connection between v_m and v while utilizing all features in V^I to encourage proximity between v_m and v. Unlike using a large number of negative samples for contrastive learning, this loss function employs positive pair comparisons, aiming to involve every feature in V^I to capture the domain-invariant semantic features of buildings. When both p_m^i and p^i are relatively large, indicating high similarity between v_m and v with v^i in V^I, the cross-entropy loss is small, which encourages the model to continue optimizing in that direction. Conversely, when both p_m^i and p^i are relatively small, indicating significant differences between v_m and v with v^i in V^I, the cross-entropy loss is large, applying a corresponding penalty to induce the model to increase the distance between v_m and v with v^i. For all features in V^I, each v^i participates fairly in the computation of the loss function, with the influence depending on the magnitude of similarity.

Additionally, the number of semantic vectors of buildings, N_v, extracted by the original model, may not be consistent with the number of selected v_m vectors, N_{v_m}, obtained through the transfer model. Therefore, it affects the value of N, which represents the number of features involved in each batch calculation. To fully utilize existing features while preventing the model from optimizing in the wrong direction, the range of N is defined as the size of N_{v_m}, and any missing part of N_v is filled by replicating the existing v vectors. If $N_v = 0$, the semantic contrastive loss function is not computed for the current batch.

4) Style Contrastive Loss: When the style feature s_d obtained from the original model has a high similarity with the

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TABLE I
DATASETS FACT SHEET

Dataset	Resolution(m)	Size(pixel)	Tiles	Area
LoveDA				
-Urban	0.3	512 × 512	5616	China
-Rural	0.3	512 × 512	3230	China
WHU	0.3	512 × 512	8189	New Zealand

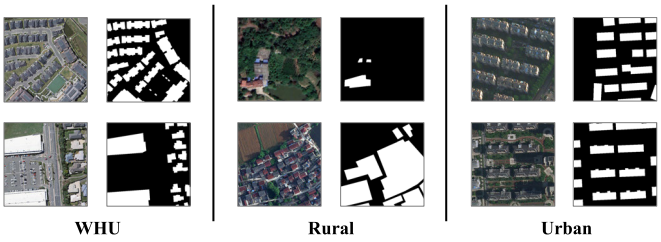


Fig. 3. Example images of datasets.

style features $S_d^S = [s_{d,1}^S, \ldots\ldots, s_{d,i}^S]$ from the same domain in the style memory bank, it is encouraged to further reduce the distance between the style features within the domain. Conversely, when s_d has a high similarity with the style features $S_{d'}^S$ from different domains in the style memory bank, it is necessary to penalize the incorrect alignment of style features across domains. To make full use of existing style features, the following contrastive loss is designed:

$$\mathcal{L}_{\text{contr}} = -\frac{1}{N_b}\sum_{\substack{d,i,j \\ d \neq d'}} \frac{\exp\left(\langle s_{d,j}, s_{d,i}^S\rangle/\tau\right)}{\exp\left(\langle s_{d,j}, s_{d,i}^S\rangle/\tau\right) + \sum_{l=1}\exp\left(\langle s_{d,j}, s_{d',l}^S\rangle/\tau\right)}, \tag{11}$$

where N_b represents the number of images per batch.

The overall loss function can be defined as follows:

$$\mathcal{L} = \mathcal{L}_{\text{Seg}} + \mathcal{L}_{\text{orth}} + \lambda_{\text{sim}}\mathcal{L}_{\text{sim}} + \lambda_{\text{contr}}\mathcal{L}_{\text{contr}}. \tag{12}$$

During the model's backpropagation update, $\mathcal{L}_{\text{Seg}}$ is used to update the feature extractor F, the semantic encoder f_t, and the decoder d. $\mathcal{L}_{\text{orth}}$ is used to update the feature extractor F, the semantic encoder f_t, and the style encoder f_s. $\mathcal{L}_{\text{sim}}$ is a constraint for semantic consistency, so it is only used to update the feature extractor F and the semantic encoder f_t. $\mathcal{L}_{\text{contr}}$ is a constraint for style consistency, so it is only used to update the feature extractor F and the style encoder f_s. λ_{sim} and λ_{contr} are the weights used to balance the $\mathcal{L}_{\text{sim}}$ and $\mathcal{L}_{\text{contr}}$, respectively, and we set them to 0.5 and 0.5, respectively.

IV. EXPERIMENTAL RESULT AND DISCUSSION

A. Datasets

To evaluate the efficacy of the proposed method, we conducted experiments on two different remote sensing datasets: LoveDA [63] and WHU Building Dataset [64]. The LoveDA dataset consists of buildings captured in two different scenes. In Table I, we compare the properties of the three building extraction datasets, and in Fig. 3, we show some sample data from these datasets.

1) LoveDA: LoveDA is a dataset specifically designed for domain adaptive semantic segmentation, featuring high spatial resolution and encompassing diverse object scales, complex scenes, and inconsistent feature distribution. The dataset comprises 0.3 m resolution images with dimensions of 1024 Í 1024 pixels, collected from various cities in China, covering a total area of 536.15 km^2. The dataset includes both urban and rural scenes, with the urban scenes characterized by a rich diversity of man-made objects in terms of styles and scales. In contrast, the rural scenes predominantly consist of natural objects, with man-made objects such as buildings belonging to the long-tail class. This difference between urban and rural scenes results in significantly different data distributions, making it challenging to transfer knowledge between these two domains for domain adaptation tasks. For building extraction, only the building labels are required, while labels for other classes can be considered as background. Given that the negative classes primarily comprise background classes, capturing building features becomes crucial for the model's performance. Consequently, data belonging to all background classes are eliminated. All images are subsequently cropped to dimensions of 512 × 512 pixels. After preprocessing, the Urban dataset consists of 5616 images, while the Rural dataset consists of 3230 images. For both datasets, 80% of the data is used for training, and the remaining 20% is used for testing.

2) WHU Building Dataset: The dataset utilized in this study is derived from high-resolution aerial imagery taken over Christchurch, New Zealand, covering an area of 450 km^2. The imagery provides a detailed view of the area with a resolution of 0.075 m and identifies approximately 220 000 individual buildings. The buildings are distributed in different scenarios, including countryside, residential, cultural, and industrial areas. To ensure computational efficiency while preserving relevant details, the original 0.075 m resolution aerial imagery is downsampled to a resolution of 0.3 m. Subsequently, the entire dataset is partitioned into 8189 smaller image patches, each measuring 512 × 512 pixels. To construct the training set, 80% of these image patches are selected, while the remaining patches are allocated for testing and evaluation purposes.

B. Implementation Details

All experiments were implemented using the PyTorch deep learning framework. The feature extractor F and decoder d were constructed using the DeepLabV3+ architecture with ResNet101 as the backbone. During the training phase, the stochastic gradient descent (SGD) optimizer with a momentum of 0.9 was employed as the optimization algorithm for the base model. The initial learning rate was set to 2e-4, and a weight decay of 4e-1 was applied. For the transfer learning model, a momentum-based update strategy was utilized

$$\omega_m = \alpha\omega_m + (1-\alpha)\omega \tag{13}$$

where ω_m represents the parameters of the transfer learning model, ω represents the parameters of the base model, and α is the momentum coefficient, set to 0.99 in our experiments. To fully demonstrate the effectiveness and superiority of our method, we conducted experiments on six tasks using

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three datasets in the comparative experiments section. The same hyperparameter configuration, image size, and batch size (set to 4) were used for each task. When selecting v_m, the confidence threshold was set to 0.85, and K was set as min(eligible, 64). The capacity of both the semantic memory bank and style memory bank was set to 256. For the overall loss function, the weight of each sub-loss function was empirically set to 1, indicating equal importance. To ensure that the model can obtain reliable building information from the supervised information in the source domain and construct a complete memory bank, $\mathcal{L}_{\text{sim}}$ and $\mathcal{L}_{\text{contr}}$ were not used to update the model in the first 1000 iterations. The comparative methods used in the experiments were all implemented with their default configurations.

C. Evaluation Metrics

For the quantitative analysis of building extraction results, $F1$-score ($F1$) and Intersect over Union (IoU) are used to evaluate the difference between the ground-truth and the predicted maps

$$F1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{14}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{15}$$

D. Result on Different Tasks

1) LoveDA-Rural as Source Dataset: Table II shows the quantitative analysis results on the other two datasets when using the Rural dataset as the source domain. The proposed method outperforms the other comparative methods on both tasks. Compared to the method that only uses the source domain dataset for model training, all domain adaptation methods achieve significant improvements in accuracy. Specifically, the AdaptSegNet [65] method, which performs domain adaptation through adversarial learning at different feature levels, achieves a 10.14% increase in $F1$-score and a 10.26% increase in IoU in the Rural ⟶ Urban task compared to the method that only uses the source domain data. In the relatively simpler Rural ⟶ WHU task, there is only a slight improvement of 1.32% in $F1$-score and 0.7% in IoU. CLAN [66] is a domain adaptation method that aligns at the category level. Compared to the method that only uses the source domain data, CLAN achieves a 7.05% increase in $F1$-score and a 7.68% increase in IoU in the Rural ⟶ Urban task. In the Rural ⟶ WHU task, there is a 6.24% improvement in $F1$-score and a 4.49% improvement in IoU. The ADVENT [67] method is an image-level domain adaptation method that mainly minimizes entropy. Compared to the method that only uses the source domain data, ADVENT achieves a 14.17% increase in $F1$-score and a 15.01% increase in IoU in the Rural ⟶ Urban task. In the Rural ⟶ WHU task, there is a 13.16% improvement in $F1$-score and a 10.19% improvement in IoU. FDANet [68] uses both image-level and feature-level strategies to perform domain adaptation tasks through adversarial learning and self-training. Compared to the aforementioned domain adaptation methods, FDANet achieves a 16.15% increase in $F1$-score and a 16.35% increase in IoU in the Rural ⟶ Urban task. In the Rural ⟶ WHU task, there is a 17.74% improvement in $F1$-score and an 18.14% improvement in IoU. ProCA [69] is an UDA method that aligns the class-centered distribution by prototypical contrast learning. Compared to the FDANet, ProCA achieves a 4.34% increase in $F1$-score and a 5.07% increase in IoU in the Rural ⟶ Urban task. In the Rural ⟶ WHU task, there is a 1.69% improvement in $F1$-score and a 3.10% improvement in IoU. MDANet [16] enhances the accurate representation of domain-invariant features for building semantics through contrastive learning and reuses domain-invariant features using a memory mechanism. By memorizing the domain-invariant features obtained from the source domain and comparing them with the features obtained in the target domain, domain shift is reduced at the feature level, enabling the transfer of key knowledge. Therefore, compared to the method that only uses the source domain, MDANet achieves significant improvements in accuracy: a 22.52% increase in $F1$-score and a 21.91% increase in IoU in the Rural ⟶ Urban task, and a 19.93% increase in $F1$-score and a 22.91% increase in IoU in the Rural ⟶ WHU task. The proposed memory optimization method based on style-semantic feature decoupling takes into account the relationship between style information and semantic information. By further optimizing domain-invariant semantic features through within-domain style consistency and intraclass object semantic consistency, the method achieves a 4.41% increase in $F1$-score and a 2.77% increase in IoU in the Rural ⟶ Urban task compared to MDANet, which also uses contrastive learning and memory mechanism. In the WHU dataset, which exhibits significant style similarities, the $F1$-score improves by 2.13% and IoU improves by 1.42% due to the relatively smaller influence of style information.

TABLE II
QUANTITATIVE ACCURACY WHEN RURAL AS SOURCE DOMAIN

Method	Rural→Urban		Rural→WHU	
	F1	IoU	F1	IoU
Source only	0.5091	0.3410	0.6013	0.4562
AdaptSegNet [65]	0.6105	0.4436	0.6145	0.4632
CLAN [66]	0.5796	0.4178	0.6637	0.5011
ADVENT [67]	0.6508	0.4911	0.7329	0.5581
FDANet [68]	0.6706	0.5045	0.7787	0.6376
ProCA [69]	0.7140	0.5552	0.7956	0.6686
MDANet [16]	0.7343	0.5601	0.8006	0.6853
DSSF(Ours)	**0.7784**	**0.5878**	**0.8219**	**0.6995**

Figs. 4 and 5 illustrate the qualitative results when the LoveDA-Rural dataset is used as the source domain, providing a visual comparison of the different methods, where part labels (a) and (b) distinguish the methods of simple domain-invariant features and further refined domain-invariant features. By observing and analyzing the imagery and attributes of the datasets, it can be noted that the Rural dataset, sampled from rural areas in various provinces and cities in China, contains a large number of natural land geo-objects. The buildings in this dataset are predominantly low-rise rural residences, with scattered and low-density distribution. On the other hand, the Urban dataset is sampled from urban areas in multiple provinces and cities in China, where artificial structures dominate the imagery, featuring diverse styles of

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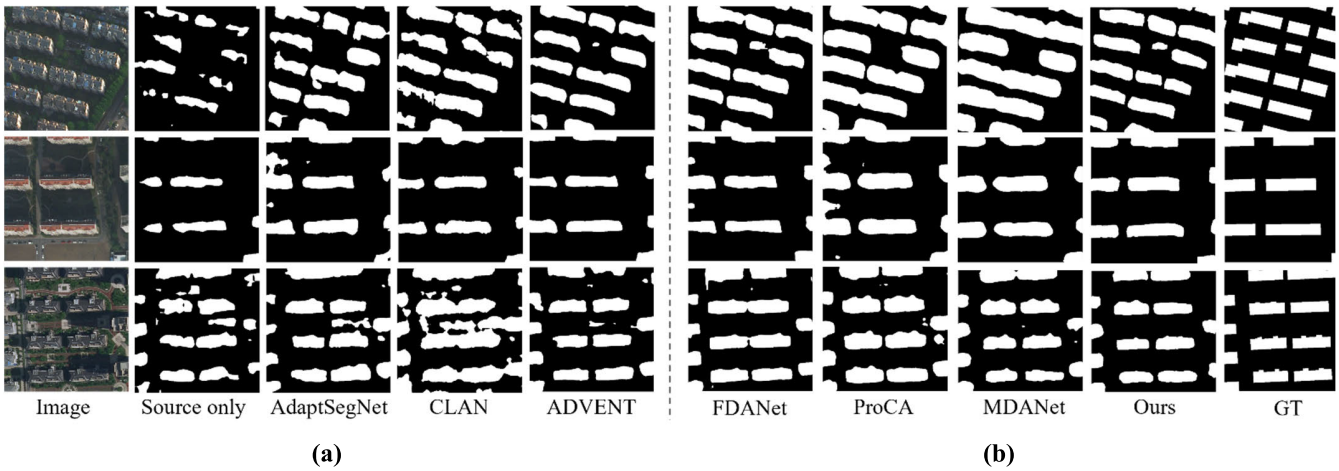


Fig. 4. Qualitative examples for the Rural→Urban case.

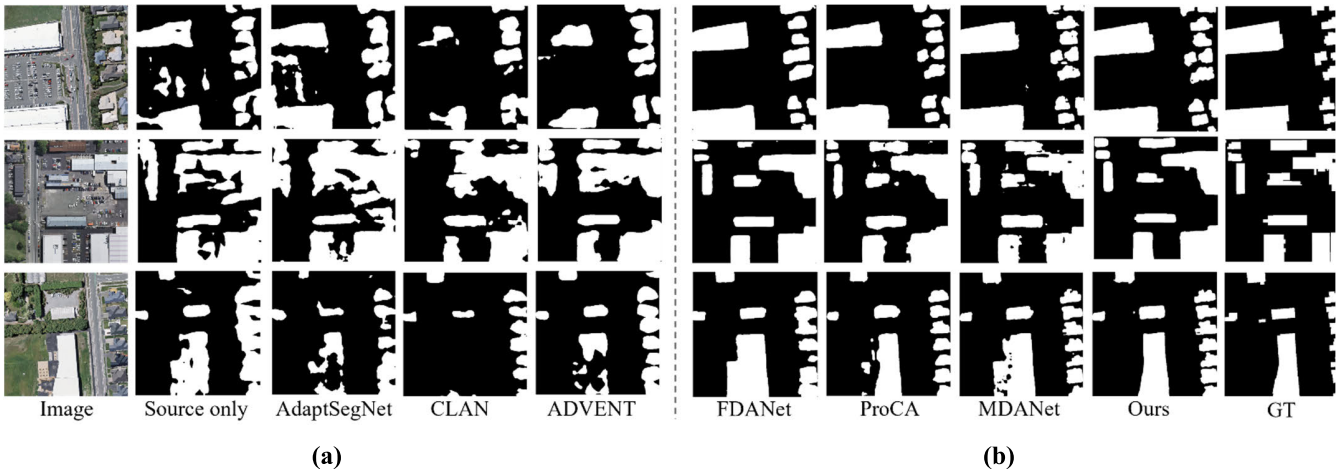


Fig. 5. Qualitative examples for the Rural→WHU case.

modern high-rise buildings. Additionally, there are significant displacements between roofs and building foundations (nonorthorectified imagery). Furthermore, the two datasets also exhibit some differences in color tones due to variations in imaging conditions.

In summary, the overall style differences between the datasets make it challenging for models to accurately capture the semantic information of buildings in the target domain, resulting in significant negative effects on cross-domain building extraction. As shown in Fig. 4(a), in the second column, when training the model using only the source domain dataset (Rural) and performing building extraction on the Urban dataset, there are numerous cases of incomplete building extraction or misidentifying other land features as buildings. However, by leveraging output labels for adversarial learning or aligning source and target domain features at the feature level, as shown in columns 3 to 5 of Fig. 4(a), it is possible to reduce missed detections or false alarms for buildings.

Nevertheless, without refining the domain-invariant features, the improvement in performance is limited. FDANet is a cross-domain building extraction method that applies strategies at both the image and feature levels. Its building extraction results, as shown in the first column of Fig. 4(b), exhibit noticeable improvements compared to the previous methods. However, there are still occurrences of false alarms and cases of connected building objects. ProCA employs contrast learning to simultaneously model intra and interclass distributions for refining the domain-invariant features. The results are shown in the second column of Fig. 4(b), which also exhibit noticeable improvements compared to the previous methods. The third column in Fig. 4(b) shows the memory contrast network proposed by the authors in a previous work, which optimizes and reuses domain-invariant features through contrastive learning and a memory mechanism, resulting in a more significant improvement in performance compared to the previous methods. However, there are still cases of missed detections since the influence of domain style information on domain-invariant features is not considered. As shown in the fourth column of Fig. 4(b), the proposed method in this article further enhances the accuracy of domain-invariant features by considering style consistency and semantic consistency, resulting in the best visual results compared to other comparative methods.

For the Rural $\longrightarrow$ WHU task, the WHU dataset, collected from European countries, exhibits characteristics of low building density and short building heights in remote sensing imagery, similar to the Rural dataset. However, there are still differences in terms of building roof materials and imaging

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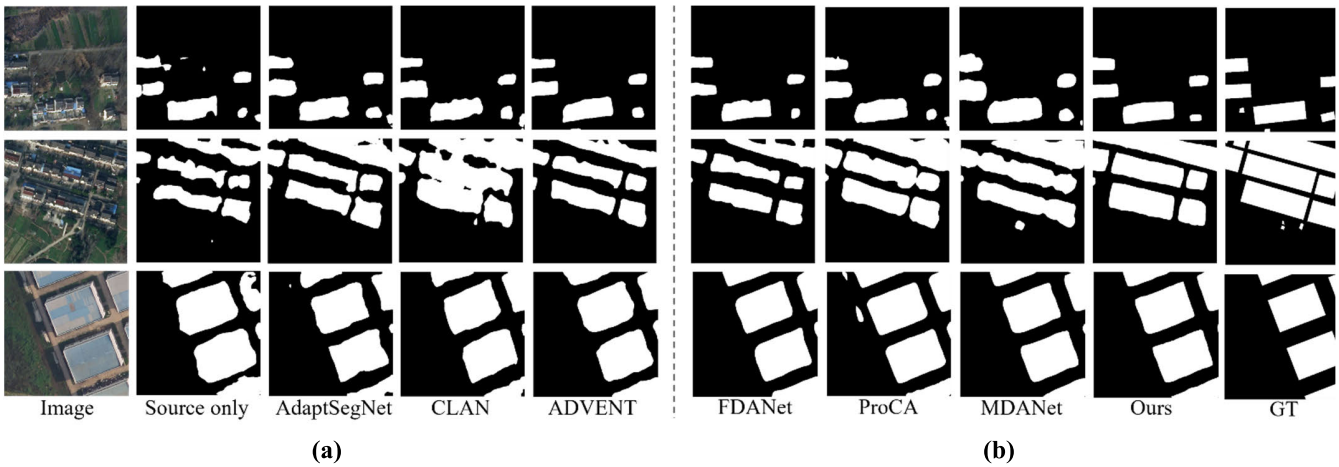


Fig. 6. Qualitative examples for the Urban→Rural case.

conditions, leading to certain style disparities between the two datasets. By observing the second column of Fig. 5(a), when training the model using only Rural data, there are almost no missed building detections, but false alarms are quite prominent. When examining columns 3 to 5 of Fig. 5(a), it can be observed that false alarms are significantly reduced, but missed detections occur. FDANet and ProCA show considerable improvement compared to the previous methods, as shown in the first and second columns of Fig. 5(b), but there are noticeable cases of connected buildings. On the other hand, MDANet, presented in the third column, reduces the occurrence of connected buildings but exhibits incomplete recognition for some building objects. The proposed method in this article achieves the best results in terms of missed detections, false alarms, and connected buildings.

2) LoveDA-Urban as Source Dataset: Fig. 6 shows the visualization results of the task using the LoveDA-Urban dataset as the source domain and the Rural dataset as the target domain. Both datasets, Urban and Rural, are collected from multiple provinces and cities in mainland China. The Urban dataset contains a diverse range of building styles, covering some of the building styles present in the Rural dataset. Overall, buildings in urban areas are predominantly high-rise structures, with a significantly lower proportion of natural land features compared to rural areas, resulting in style differences between the two datasets. As shown in the second column of Fig. 6(a), when training the model using only the Urban dataset, although it can generally recognize buildings, the extracted building boundaries are blurry, and there are noticeable cases of incomplete building extraction. As shown in columns 3 to 5 of Fig. 6(a) and columns 1 to 3 of Fig. 6(b), using different domain adaptation methods can improve the building extraction performance, consistent with the quantitative results presented in Table III. However, these methods lack stability and cannot consistently achieve satisfactory results on all test data, as shown in the first row of Fig. 6(a) and (b), where these methods exhibit varying degrees of boundary blurring and object connections. The proposed method in this article shows significantly better performance in building boundary recognition compared to other comparative methods, although there are still some missed detections of small building objects. However, it achieves the best accuracy, as shown in Table III.

TABLE III
QUANTITATIVE ACCURACY WHEN URBAN AS SOURCE DOMAIN

Method	Urban→Rural		Urban→WHU	
	F1	IoU	F1	IoU
Source only	0.6207	0.4541	0.5841	0.4216
AdaptSegNet [65]	0.7417	0.5669	0.6633	0.5001
CLAN [66]	0.7041	0.5298	0.6502	0.4909
ADVENT [67]	0.7245	0.5497	0.6772	0.5155
FDANet [68]	0.7351	0.5598	0.7263	0.5503
ProCA [69]	0.7571	0.6092	0.7702	0.6207
MDANet [16]	0.7813	0.6171	0.7671	0.6048
DSSF(Ours)	**0.7906**	**0.6285**	**0.7964**	**0.6423**

The visualization results for the Urban ⟶ WHU task are shown in Fig. 7. When trained using only the source domain dataset, the model fails to capture the semantic information of buildings in the target domain, resulting in numerous missed detections and false alarms. The WHU dataset is sampled from European countries and exhibits significant differences compared to the Urban dataset sampled from Chinese provinces and cities. The buildings in the WHU dataset are mostly low-rise, standalone structures, and there are noticeable differences in image color tones due to imaging conditions. As shown in Fig. 7, using domain adaptation strategies consistently achieves better results compared to using only the source domain method. The proposed method in this article not only reduces missed detection and false alarm rates but also produces smoother building boundaries. Table III also confirms the superiority of our method over other comparative methods.

3) WHU as Source Dataset: Fig. 8 shows the visualization results of the task using the WHU dataset as the source domain and the Rural dataset as the target domain. By observing the first row of the second column in Fig. 8(a), it can be seen that when training the model using only the WHU dataset, the model mistakenly identifies roads as buildings. This is because the style information greatly influences the model's expression of building semantics. Even in the domain adaptation methods shown in columns 3 to 5 of Fig. 8(a) and

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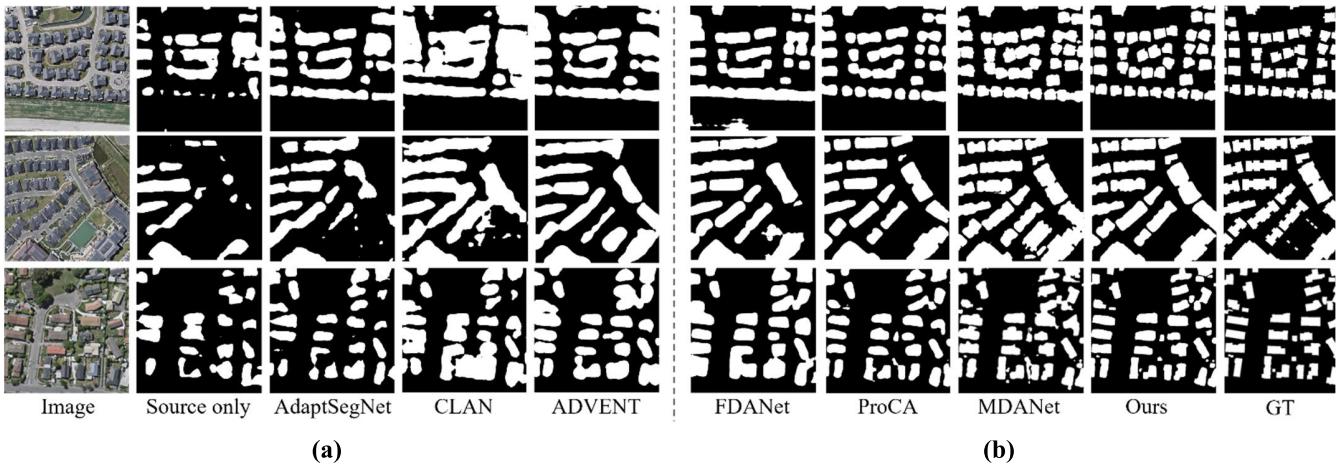


Fig. 7. Qualitative examples for the Urban→WHU case.

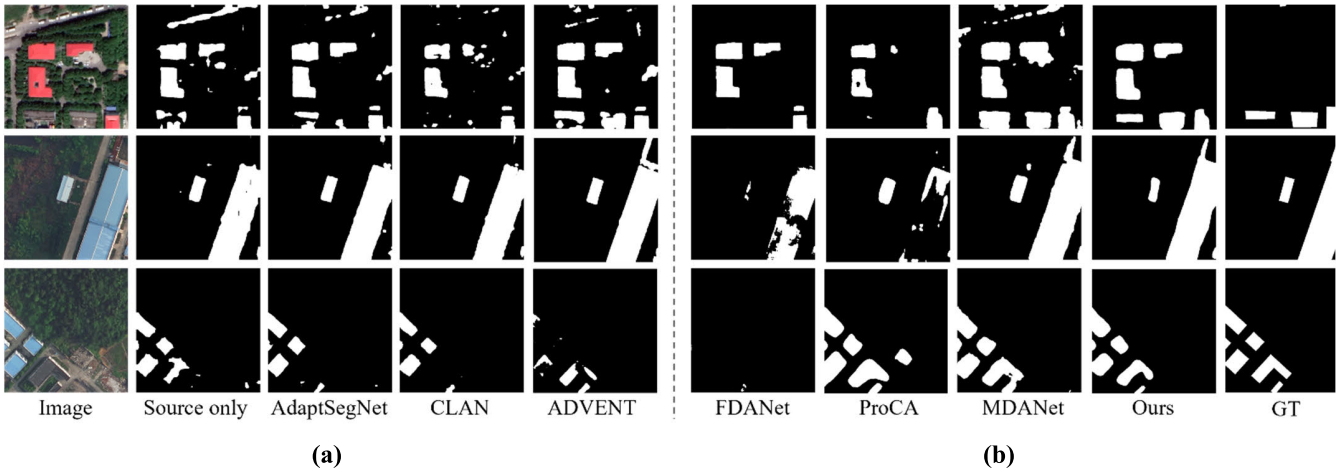


Fig. 8. Qualitative examples for the WHU→Rural case.

columns 1 to 3 of Fig. 8(b), only the FDANet method does not confuse roads with buildings. However, the FDANet method exhibits unstable performance and does not yield satisfactory results in the test cases shown in the second and third rows of Fig. 8(a) and (b). By observing the first and last columns of Fig. 8, it can be observed that in the example of the first row, there are some errors in the labels, and the building objects are not fully annotated. However, all comparative methods can partially or completely extract these buildings, but they all exhibit noticeable missed detection of buildings. Only the proposed method in this article can not only extract the unmarked buildings in the labels but also ensure the accuracy of detected labeled buildings. Table IV also reflects that the proposed method in this article achieves the best quantitative results.

The Urban dataset contains a large number of high-rise buildings, and due to the nonorthorectified nature of the target domain images, a noticeable displacement phenomenon occurs where the roofs and building facades appear simultaneously in the images. Therefore, when the Urban dataset is used as the target domain, all the methods shown in columns 2 to 5 of Fig. 9(a) and the 1 to 2 column of Fig. 9(b) exhibit varying degrees of missed detection. The building

TABLE IV
QUANTITATIVE ACCURACY WHEN WHU AS SOURCE DOMAIN

Method	WHU→Urban		WHU→Rural	
	F1	IoU	F1	IoU
Source only	0.3392	0.2009	0.4717	0.3104
AdaptSegNet [65]	0.4627	0.3146	0.4898	0.3243
CLAN [66]	0.4695	0.3086	0.5058	0.3376
ADVENT [67]	0.4511	0.2954	0.4731	0.3129
FDANet [68]	0.6290	0.4778	0.5102	0.3424
ProCA [69]	0.6898	0.5153	0.5989	0.4275
MDANet [16]	0.7277	0.5507	0.7205	0.5465
DSSF(Ours)	**0.7713**	**0.6114**	**0.7471**	**0.5726**

extraction performance of MDANet is significantly better than the aforementioned methods, while the proposed method in this article achieves smoother handling of building boundaries and reduces the occurrence of object fusion. The quantitative evaluation shown in Table IV demonstrates that the proposed method in this article achieves the best results.

Table V provides statistics on the model parameters and floating-point operations (FLOPs) used by the domain adaptation methods. The proposed method in this article has

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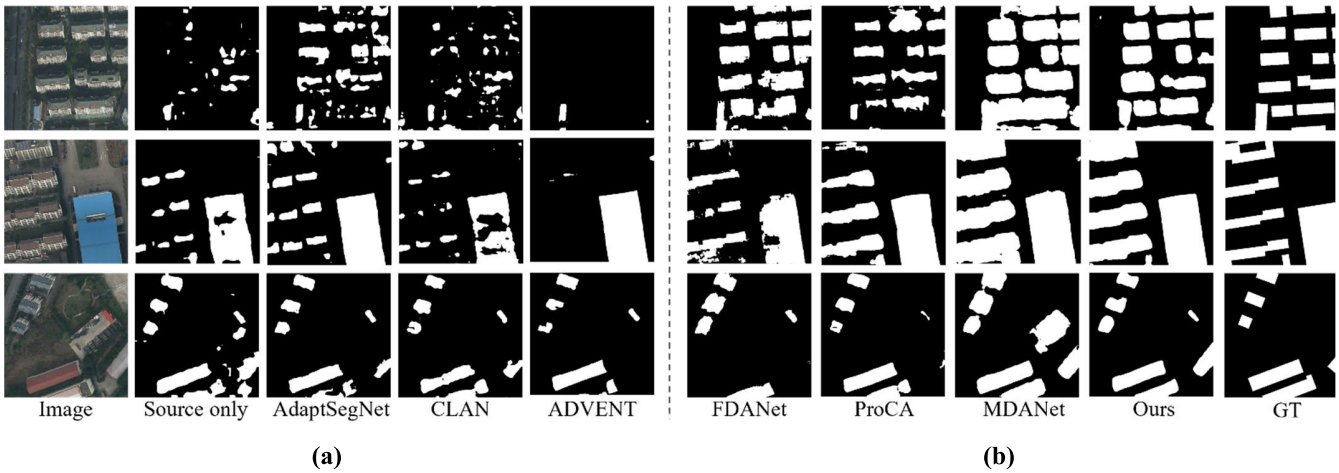


Fig. 9. Qualitative examples for the WHU→Urban case.

TABLE V
MODEL PARAMETERS AND TRAINING EFFICIENCY

Method	AdaptSegNet	CLAN	ADVENT	FDANet	ProCA	MDANet	Ours
Parameters(M)	186	166	186	79	43	108	101
FLOPs (GMac)	186	193	186	137	179	201	235

a relatively low parameter count due to the absence of parameter learning for memory updates. While it has higher computational complexity due to the introduction of multiple contrastive losses, it achieves optimal performance in terms of accuracy for each task. Although the FDANet method has the lowest computational requirements and the ProCA method has the lowest parameters, their performance in cross-domain building extraction is not stable. Therefore, the computational cost of our method is justified considering its performance and the lower parameter count.

E. Discussion

1) Ablation Study: The effectiveness of this method has been proven through qualitative and quantitative analysis of experimental results. To further validate the positive effects of style consistency and semantic consistency constraints in the decoupling of style-semantic features and optimization of memory features, ablation experiments were conducted in this section. Specifically, the orthogonal constraint loss function $\mathcal{L}_{\text{orth}}$, semantic contrastive loss function $\mathcal{L}_{\text{sim}}$, and style contrastive loss function $\mathcal{L}_{\text{contr}}$ were quantitatively analyzed as variables, as shown in Tables VI and VII.

When all three losses were not used, only the semantic segmentation loss $\mathcal{L}_{\text{Seg}}$ was employed. To demonstrate the performance of each component in tasks with different style differences, these experiments were conducted on the Urban → Rural task and the WHU → Urban task. Figs. 10(a) and (b) and 11(a) and (b), show the feature response maps, where warmer colors indicate a higher response to buildings.

When all three losses were not used and only the semantic segmentation loss $\mathcal{L}_{\text{Seg}}$ was employed, according to Table VII, it can be observed that the accuracy of the selected tasks in this experiment was comparable to the accuracy achieved by training the model using only the source domain data. However, when only the orthogonal constraint loss function $\mathcal{L}_{\text{contr}}$ and the semantic segmentation loss $\mathcal{L}_{\text{Seg}}$ were used,

TABLE VI
RESULTS OF THE ABLATION STUDIES ON THE URBAN→RURAL TASK

$\mathcal{L}_{orth}$	$\mathcal{L}_{sim}$	$\mathcal{L}_{contr}$	F1	IoU
	√		0.6223	0.4534
√			0.5934	0.4297
	√		0.7226	0.5493
√		√	0.6984	0.5327
√	√		0.7650	0.5899
√	√	√	**0.7906**	**0.6285**

TABLE VII
RESULTS OF THE ABLATION STUDIES ON THE WHU→RURAL TASK

$\mathcal{L}_{orth}$	$\mathcal{L}_{sim}$	$\mathcal{L}_{contr}$	F1	IoU
	√		0.3503	0.2137
√			0.3162	0.1808
	√		0.6730	0.5261
√		√	0.5453	0.3801
√	√		0.7203	0.5468
√	√	√	**0.7713**	**0.6114**

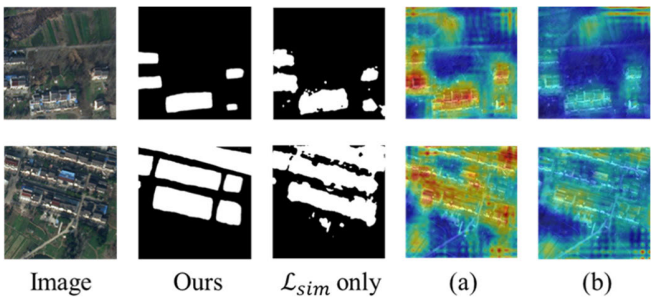


Fig. 10. Visualization results of semantic feature responses on the Urban → Rural task are shown below. (a) Semantic feature response maps of the proposed method in this article. (b) Semantic feature response maps when only the semantic contrastive loss is used.

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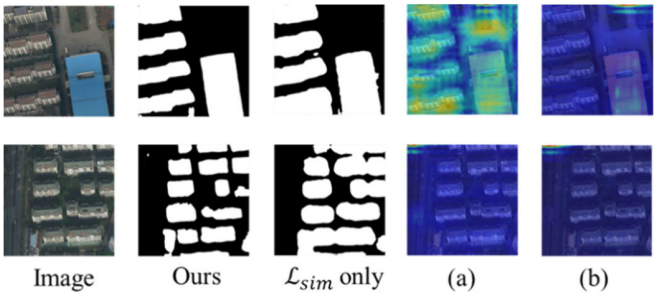


Fig. 11. Visualization results of semantic feature responses on the WHU⟶Urban task are shown below. (a) Semantic feature response maps of the proposed method in this article. (b) Semantic feature response maps when only the semantic contrastive loss is used.

a significant decrease in accuracy was observed for both tasks. Although the original features were decoupled and the semantic features were constrained, the orthogonal constraint loss was applied to the original model, and the only reliable supervision information available was from the semantic segmentation loss $\mathcal{L}_{\text{Seg}}$ applied to the source domain data. Under the influence of the orthogonal constraint loss, the model leaned more toward the source domain data, making it more difficult to effectively extract features in the target domain.

When only the semantic contrastive loss function $\mathcal{L}_{\text{sim}}$ and the semantic segmentation loss $\mathcal{L}_{\text{Seg}}$ were used, a significant improvement in accuracy was observed. The $F1$-score and IoU increased by more than 10% in the Urban → Rural task and by more than 20% in the WHU → Urban task. At this point, the model updated itself by continuously narrowing the gap between the source and target domain features. However, due to the interference of style information, the alignment was limited. By observing Figs. 10(b) and 11(b), it can be noticed that the response to the semantics of buildings was relatively low, resulting in a significant amount of boundary leakage and false detections in the predicted images.

When only the semantic contrastive loss function $\mathcal{L}_{\text{sim}}$ was not used, the model benefited from the within-domain style consistency constraints, leading to improved building extraction accuracy in both tasks. Style features can be optimized through style consistency constraints, while the orthogonal constraint loss can enhance the representation of semantic features using style features. However, semantic features still dominate the model's extraction of buildings, and without semantic consistency constraints, the accuracy of semantic features is insufficient. When only the style contrastive loss function $\mathcal{L}_{\text{contr}}$ was not used, it encourages the model to capture domain-invariant semantic features and eliminate the influence of style information on semantic features through the orthogonal constraint loss. However, due to the lack of style consistency constraints, the orthogonal constraint loss cannot fully exert its intended effect. When all loss functions are used simultaneously, the synergy between semantic consistency and style consistency enhances the expression of semantic features through style features, significantly improving the transferability of domain-invariant features, and achieving the highest accuracy. By observing Figs. 10(a) and 11(a), it can be seen that the category response centers are relatively clear, with lower responses to nonbuilding areas, indicating that the model is more confident in classifying clustered regions as building categories. It is worth noting that in the example of the second row in Fig. 11, the responses of both methods are relatively low, but the overall response level of the complete method is higher compared to using only the semantic contrastive loss method.

2) Parameter Analysis: This study first conducted a parameter sensitivity analysis on the weight of style consistency loss λ_{contr} and semantic consistency loss λ_{sim}. The effectiveness of the modules revealed that semantic consistency constraint plays a dominant role, while style consistency contributes as a supportive factor. Therefore, this section analyzes the synergistic effect of the two consistency losses, with the weight of semantic consistency set greater than or equal to the weight of style consistency. The experiments were conducted on the Urban → Rural and WHU → Urban tasks.

The experimental results, as shown in Table VIII, indicate that when the weight of semantic consistency loss is fixed, the IoU increases with the increase in the weight of style consistency loss on both tasks. Moreover, a higher weight of style consistency loss leads to a more significant improvement in IoU. On the other hand, when the weight of style consistency loss is fixed, IoU also increases with the increase in the weight of semantic consistency loss, but the improvement becomes smaller as the weight of semantic consistency loss increases. The best accuracy is achieved when both weights are equal and set to 1. In summary, when the weight of semantic consistency loss is much larger than the weight of style consistency loss, the style consistency constraint is suppressed, and its auxiliary effect on semantic consistency is weakened.

The study also analyzed the impact of memory bank capacity on the model. Clearly, a larger memory bank capacity allows more features to participate in the contrastive loss, which also increases the computational cost. Therefore, it is necessary to set an appropriate capacity that balances high building extraction accuracy and relatively low computational cost. As shown in Tables IX and X, when the memory bank capacity is increased from 128 to 256, the $F1$-score and IoU in the Urban → Rural task improved by 1.48% and 2.53% respectively, and in the WHU → Urban task, they improved by 2.84% and 3.87%, respectively. This is because the expansion of the memory bank increases the constraint on semantic and style consistency in the contrastive loss, thereby improving the accuracy of domain-invariant features. In the Urban → Rural task, further increasing the memory bank capacity from 256 to 512 resulted in minimal improvement in accuracy, while in the WHU → Urban task, there was a slight decrease in accuracy. The model forces the extracted features to move along the common center of the features in the memory bank. Although the model can extract domain-invariant features with higher transferability during training, it lacks transferability in the early stages of training, indicating that excessive biased features can affect model updates. Additionally, when the capacity increases, more features satisfying the confidence requirement but with relatively low confidence are stored,

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TABLE VIII
PARAMETER SENSITIVITY ANALYSIS OF THE WEIGHTS FOR STYLE CONSISTENCY LOSS AND SEMANTIC CONSISTENCY LOSS

Task（$\lambda_{sim}/\lambda_{contr}$）	0.1/0.1	0.5/0.1	1/0.1	0.5/0.5	1/0.5	**1/1**
U-R	0.4731	0.5271	0.5663	0.5394	0.5873	**0.6285**
W-U	0.3976	0.5065	0.5527	0.5203	0.5742	**0.6114**

TABLE IX
RESULT ON DIFFERENT CAPACITY OF MEMORY BANK

Size	128	256	**512**
F1	0.7758	0.7906	**0.7984**
IoU	0.6032	0.6285	**0.6377**

TABLE X
RESULT ON DIFFERENT CAPACITY OF MEMORY BANK

Size	128	**256**	512
F1	0.7429	**0.7713**	0.7662
IoU	0.5724	**0.6114**	0.5960

which also has a negative impact on the model. Therefore, this article selects a memory bank capacity of 256, which achieves a good balance between computational cost and effectiveness.

V. CONCLUSION

This study proposes a cross-domain extraction network for buildings based on style-semantic feature decoupling, aiming to optimize the domain-invariant semantic features of buildings through the deep exploitation of intradomain style information. The model employs intradomain style consistency constraints for domain style information and semantic consistency constraints for building semantic information. By leveraging the memory mechanism and contrastive learning strategy, the model fully exploits self-supervised information to reduce the differences in domain style features and intraclass-building semantic features. To ensure the independence of style and semantic features, a style-semantic feature decoupling module is designed to utilize style features to promote accurate representation of semantic features through linearly independent orthogonal constraints. Experimental results show that the proposed method achieves outstanding building extraction performance in six cross-domain tasks. Compared to the comparative methods, this method exhibits significant advantages in stability and fine-detail handling capability.

In future work, we will continue to explore the relationship between memory mechanism and domain adaptation tasks and attempt to use more efficient strategies to optimize the domain adaptation model.

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Jie Chen (Member, IEEE) received the M.S. degree from the East China University of Technology, Nanchang, China, in 2006, and the Ph.D. degree from the Central South University, Changsha, China, in 2010.

He is now a Professor with the Department of Geo-Informatics, Central South University. His research interests include intelligent analysis and understanding of remote-sensing images.

Dr. Chen is a reviewer for several journals in the field of remote sensing.



Jingru Zhu received the B.S. degree from Shandong University of Science and Technology, Qingdao, Shandong, China, in 2018, and the M.S. degree from the Central South University, Changsha, Hunan, China, in 2021, where she is currently pursuing the Ph.D. degree.

Her research interests include remote sensing image intelligent interpretation and cross-domain semantic segmentation of high-resolution remote sensing images.



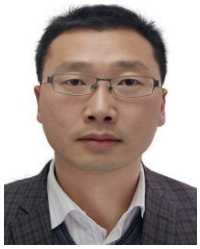
Peien He received the B.S. degree from China University of Mining and Technology, Xuzhou, Jiangsu, China, in 2020, and the M.S. degree from the Central South University, Changsha, Hunan, China, in 2023.

His research interests include remote sensing image intelligent interpretation and building cross-domain extraction of high-resolution remote sensing images.



Ya Guo received the M.S. degree from China University of Geosciences in Beijing, Beijing, China, in 2020. He is currently pursuing the Ph.D. degree with the Central South University, Changsha, China.

His research interests include remote sensing image intelligent interpretation and artificial intelligence.



Liang Hong received the M.S. degree from Kunming University of Technology, Kunming, China, in 2007, and the Ph.D. degree from Wuhan University, Wuhan, China, in 2010.

He is now a Professor and Doctoral Supervisor at the Department of Geography, Yunnan Normal University, Kunming. His current research interests include remote sensing big data intelligence processing and spatiotemporal analysis.



Yin Yang received the M.S. and Ph.D. degrees from Xiangtan University, Xiangtan, China, in 2006 and 2010, respectively.

She is now a Professor and Doctoral Supervisor at the School of Mathematics and Computational Science, Xiangtan University, Xiangtan, China. Her current research interests include numerical solutions of partial differential equations and applications.



Min Deng (Member, IEEE) received the Ph.D. degrees from Wuhan University, Wuhan, China, in 2003, and Asian Institute of Technology, Pathum Thani, Thailand, in 2004.

He is a Full Professor in cartography, GIS, and remote sensing at the Department of Geo-Informatics, Central South University, Changsha, China, where he is an Associate Dean and a Doctoral Supervisor with the School of Geosciences and Info-Physics. He has published over 160 journal articles and three authored research monographs. His current research interests include geospatial data updating, spatiotemporal data mining, and spatiotemporal analysis and modeling.



Geng Sun received the B.S. degree from the Central South University, Changsha, Hunan, China, in 2021, where he is currently pursuing the Ph.D. degree.

His research interests include intelligent interpretation of high-resolution remote sensing images, urban visual perception, and intelligent evaluation.

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